The Total Rank Formula and Image Basis Conjecture for the Staircase Product

Clio

April 16, 2026

Abstract

We reduce the H-invariant theorem $\operatorname{rank}(\Pi|_{V_\lambda}) = \dim(V_\lambda^H)$ to a single statement: the *image basis conjecture*, which asserts that right multiplication by the staircase product $\Pi \in \mathbb{Q}[S_n]$ sends the descent-paired basis elements to linearly independent vectors. Combined with the eigenvalue positivity theorem (proved via Kazhdan–Lusztig positivity), this yields the full theorem. The image basis conjecture is verified computationally for $n \leq 7$.

1 Setup and statement

Let $\Pi = \prod_{k=1}^{n-1} \prod_{j=k}^{1} (1 + s_j) \in \mathbb{Q}[S_n]$ be the staircase product, where $s_j = (j, j+1)$ is the simple transposition. Let $H = \langle s_1, s_3, s_5, \ldots \rangle \cong (\mathbb{Z}/2)^{\lfloor n/2 \rfloor}$.

Definition 1. A permutation $w \in S_n$ is *descent-paired* if $w(2i-1) > w(2i)$ for all $1 \leq i \leq \lfloor n/2 \rfloor$. Let $D \subset S_n$ denote the set of descent-paired permutations.

Proposition 2. *$|D| = n!/2^{\lfloor n/2 \rfloor}$, and D is a complete set of representatives for the right cosets of H in S_n.*

Conjecture 3 (Image Basis Conjecture)**.** *The vectors $\{e_w \cdot \Pi : w \in D\}$ are linearly independent in $\mathbb{Q}[S_n]$.*

Theorem 4 (Total Rank Formula)**.** *If Conjecture **??** holds for n, then*

$$\operatorname{rank}(\Pi \text{ on } \mathbb{Q}[S_n]) = \frac{n!}{2^{\lfloor n/2 \rfloor}}.$$

Theorem 5 (H-Invariant Theorem)**.** *Assuming Conjecture **??** for n: for every partition $\lambda \vdash n$,*

$$\operatorname{rank}(\Pi|_{V_\lambda}) = \dim(V_\lambda^H).$$

2 Proofs

2.1 Positive semidefiniteness of the convolution form

Write $\Pi = \sum_{g \in S_n} f_g \, g$ where $f_g > 0$ for all g (since every permutation appears with positive coefficient in the staircase product, being the evaluation of a Schubert polynomial at 1).

Lemma 6. *The bilinear form $B(x, y) = \langle x, R_\Pi\, y\rangle$ on $\mathbb{Q}[S_n]$ is positive semidefinite, where R_Π denotes right multiplication by Π and $\langle\cdot,\cdot\rangle$ is the standard inner product on the group algebra (with $\{e_w\}$ orthonormal).*

Proof. The right multiplication operator R_Π on the regular representation decomposes as $\bigoplus_\lambda \mathrm{Id}_{V_\lambda} \otimes \rho_{\lambda^*}(\Pi)$, where λ^* is the conjugate partition. By the eigenvalue positivity theorem [**?**], all nonzero eigenvalues of $\rho_\mu(\Pi)$ are positive for every μ. Hence R_Π is positive semidefinite. □

Lemma 7. *$f_g = f_{g^{-1}}$ for all $g \in S_n$. Hence the matrix $M_R[v, w] = f_{w^{-1}v}$ is symmetric.*

Proof. The staircase reduced word $\mathbf{i} = (i_1, \ldots, i_m)$ has the property that its reversal $(i_m, \ldots, i_1)$ is also a reduced word for w_0. A subset $J \subseteq \{1, \ldots, m\}$ gives a reduced expression for w in the forward reading if and only if $\{m+1-j : j \in J\}$ gives a reduced expression for w^{-1} in the reverse reading. Since Π counts subwords of the staircase word, $f_g = f_{g^{-1}}$. (Alternatively, this is verified computationally for all $n \le 7$.) □

2.2 Reduction to the image basis conjecture

Proof of Theorem **??**. By Lemma **??**, R_Π is positive semidefinite. The submatrix $M_R[D, D]$ with entries $f_{w^{-1}v}$ for $v, w \in D$ is a principal submatrix of a PSD matrix, hence PSD.

Conjecture **??** states that the columns of the right multiplication matrix R_Π indexed by D are linearly independent. This holds if and only if $M_R[D, D]$ is positive definite (since for a PSD form, the Gram matrix of a subset of vectors is PD iff the vectors are linearly independent).

If $M_R[D, D]$ is PD, then $\mathrm{rank}(R_\Pi) \ge |D| = n!/2^{\lfloor n/2\rfloor}$.

For the upper bound: $\Pi \cdot s_1 = \Pi$ (the rightmost factor of each block is $(1 + s_1)$, which absorbs s_1). Hence $R_\Pi(e_w - e_{s_1 w}) = 0$ for all w, giving $\mathrm{rank}(R_\Pi) \le n!/2$. More generally, the cascade analysis [**?**] yields the per-irrep upper bound $\mathrm{rank}(\Pi|_{V_\lambda}) \le \dim(V_\lambda^H)$ via the eigenvalue positivity theorem. Summing over irreducibles using the regular representation decomposition:

$$\mathrm{rank}(R_\Pi) = \sum_\lambda \dim(V_\lambda) \cdot \mathrm{rank}(\Pi|_{V_\lambda}) \le \sum_\lambda \dim(V_\lambda) \cdot \dim(V_\lambda^H) = \frac{n!}{|H|} = \frac{n!}{2^{\lfloor n/2\rfloor}}.$$

The last equality is Frobenius reciprocity applied to $\mathrm{Ind}_H^{S_n}(\mathbf{1})$. □

Proof of Theorem **??**. By Theorem **??**:

$$\sum_\lambda \dim(V_\lambda) \cdot \mathrm{rank}(\Pi|_{V_\lambda}) = \frac{n!}{2^{\lfloor n/2\rfloor}} = \sum_\lambda \dim(V_\lambda) \cdot \dim(V_\lambda^H).$$

Since $\mathrm{rank}(\Pi|_{V_\lambda}) \le \dim(V_\lambda^H)$ for every λ (eigenvalue positivity + upper bound comparison [**?**]), equality of the weighted sums forces $\mathrm{rank}(\Pi|_{V_\lambda}) = \dim(V_\lambda^H)$ for every λ. □

3 Computational verification

The image basis conjecture is equivalent to positive definiteness of $M_R[D, D]$. We verify this by computing exact determinants:

n	$\lvert D\rvert$	$\det(M_R[D, D])$	Factored	Min eigenvalue	PD?
3	3	4	2^2	1.000	Yes
4	6	576	$2^6 \cdot 3^2$	0.658	Yes
5	30	$2^{70} \cdot 3^{10}$	—	0.693	Yes
6	90	$2^{272} \cdot 3^{52} \cdot 5^8 \cdot 7^2 \cdot 11^8$	—	0.551	Yes
7	630	$> 10^{300}$	—	0.584	Yes

Key structural properties of $M_R[D, D]$:

- Symmetric ($f_g = f_{g^{-1}}$).
- Constant diagonal: $M_R[w, w] = f_e$ for all $w \in D$.
- Positive definite for all tested n.
- The minimum eigenvalue appears to remain bounded away from 0 (between 0.5 and 1.0 for all $n \leq 7$).

4 Discussion

Why right multiplication. The left multiplication submatrix $M_L[D, D]$ (with entries $f_{vw^{-1}}$) is singular for $n \geq 3$: distinct descent-paired permutations can yield identical rows. The right multiplication submatrix $M_R[D, D]$ (with entries $f_{w^{-1}v}$) avoids this degeneracy and is positive definite. The crucial asymmetry arises because $f_{vw^{-1}} \neq f_{w^{-1}v}$ in a non-abelian group.

Structure of the determinant. The determinants factor into powers of small primes, suggesting a product formula. The diagonal value f_e (the coefficient of the identity in Π, counting subwords of the staircase giving e) satisfies $f_e = 2, 6, 34, 342, 6408$ for $n = 3, 4, 5, 6, 7$.

Proof status. The full H-invariant theorem (Theorem **??**) is **proved** modulo Conjecture **??**. All other ingredients—eigenvalue positivity, the upper bound, the Frobenius squeeze—are established unconditionally. Conjecture **??** is verified through $n = 7$ ($|D| = 630$, exact integer determinant nonzero).

Possible proof strategies for Conjecture ??.

1. **Cascade induction**: Track linear independence of $\{e_w \cdot \Pi_{\leq k} : w \in D\}$ through the staircase factors, showing independence is preserved at each step.
2. **Total positivity**: The factored structure of Π as a product of elementary "row addition" maps may yield total nonnegativity of $M_R[D, D]$, implying positive definiteness.
3. **Representation-theoretic**: Use the Kazhdan–Lusztig cell structure of the Hecke algebra to control the rank of Π on parabolic quotients.

References

[Clio-EP] Clio, *Eigenvalue positivity for the Hecke staircase element via Kazhdan–Lusztig positivity*, April 2026.

[Clio-cascade] Clio, *Cascade proof of the H-invariant upper bound*, April 2026.